AI & Software Engineering Workshop

Week 2, Day 4: Training, Part 2

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Morning check-in

- How many steps did it cover overnight?
- Loss: where did it start yesterday, where is it now?
- Generate from the latest checkpoint: read it out loud
- Compare with yesterday's step-1000 sample

It studied while you slept. Let's see the homework.

What's inside the box

- **Embeddings**: every token becomes a vector (a point in meaning-space)
- **Attention**: each word looks back: "*which earlier words matter to me?*"
- **Feed-forward layers**: where pattern-knowledge is stored
- ×24 layers deep: each layer refines the previous one's understanding
- 350M parameters = all the numbers in those layers, learned from data

Reading the loss curve

- **Healthy**: falls fast, then keeps grinding down slowly
- **Underfitting**: flat and high, model too small for the data
- **Overfitting**: training loss falls, *validation* loss rises, memorizing not learning
- Our setup: 350M params vs 8.3B tokens, plenty of data per parameter

The curve is the model's EKG. Learn to read it.

Temperature: one model, many personalities

```
python 5_generate.py --prompt "Երևանը" --temperature 0.3
python 5_generate.py --prompt "Երևանը" --temperature 1.0
```

- **Low**: confident, repetitive, safe
- **High**: creative, surprising, occasionally unhinged
- Same weights. Only the *sampling* changed.

Find your model's sweet spot; it's usually ~0.6–0.8.

Measure, don't vibe

Two scorecards in the repo:

- `eval_bpb.py`: bits-per-byte on held-out text, how well does it predict Armenian it has *never seen*?
- `eval_armbench.py`: Armenian multiple-choice questions, does it *know* things?

Score an early checkpoint vs. the final one. Numbers, not impressions: that's how you know a change helped.

The finish line

- Training completes → `checkpoints/final.pt`
- Generate with the final model, then with Day 1's tiny model
- Same prompt, side by side. Read both aloud.
- **That gap is what 8 hours and 350M parameters buy.**

Today → Tomorrow

Today: a trained Armenian base model that continues any text you give it.

Tomorrow: it learns to *answer* instead of just continue, then we ship it and wire it into your Week 1 bot.